

SEARCHING PROBLEM 1

QUESTION:

DS	Session	Searching	Question Information	Level 1	Challenge 1
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Problem Description:
Mr. Alex just installed new bands in his feet and they do not feel like they should, making it where he cannot stop!

However the Alex cannot do the math while stopping himself from crashing into things. Using $M = -dx$, please help him find out the M, d, or x value as he requests it.

Constraints:
M, D, X = 0 to 10000 (or a question mark).

Input Format:
Each line will start capital letters only, then the value of that variable. The value must be a decimal number.

Output Format:
Print the output in a separate lines contains the variable, a space, then the value.

Logical Test Cases

Test Case 1 INPUT (STDIN) M 14.00 D -4.80 X ? EXPECTED OUTPUT x 2.92	Test Case 2 INPUT (STDIN) M 12.00 D -5.80 X ? EXPECTED OUTPUT x 2.07
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Mandatory Test Cases

Test Case 1 KEYWORD char var[3][LEN];	Test Case 2 KEYWORD char inp[3][LEN];
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Complexity Test Cases

Test Case 1 CYCLOMATIC COMPLEXITY 4	Test Case 2 TOKEN COUNT 320	Test Case 3 NLOC 30
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SOURCE CODE:

```
#include <stdio.h>
#include <string.h>
#include <math.h>

int main() {
    double mass = 0.0, dist = 0.0, pos = 0.0;
    int find_m = 0, find_d = 0, find_x = 0;

    for (int k = 0; k < 3; k++) {
        char symbol[10], value_str[30];
        if (scanf("%s %s", symbol, value_str) != 2) return 0;

        if (strcmp(value_str, "?") == 0) {
            if (symbol[0] == 'M') find_m = 1;
            else if (symbol[0] == 'D') find_d = 1;
            else if (symbol[0] == 'X') find_x = 1;
        } else {
            double parsed;
            sscanf(value_str, "%lf", &parsed);
            if (symbol[0] == 'M') mass = parsed;
            else if (symbol[0] == 'D') dist = parsed;
            else if (symbol[0] == 'X') pos = parsed;
        }
    }

    if (find_x) {
        printf("x %.2f\n", fabs(mass / dist));
    } else if (find_d) {
        printf("d %.2f\n", fabs(mass / pos));
    } else if (find_m) {
        printf("m %.2f\n", fabs(dist * pos));
    }
    return 0;
}
```

OUTPUT:

```
M 14.00  
D -4.80  
X ?  
x 2.92
```

```
M 12.00  
D -5.80  
X ?  
x 2.07
```